Start here: what you are building

This guide explains the choices, the commands, the phone setup and how to prove the result works. Read the introduction and sections 1-2 first, then follow the numbered stages in order.

NekoServer installs and connects a collection of server applications on an **already installed Arch/Nyarch Linux system**. It does not install the operating system from a USB stick or format your disks. It is intended for a fresh setup; existing unrelated containers and data may require separate migration.

Service	What you use it for	Where the main files live
Immich	Your phone photos and videos	Application SSD
Jellyfin	Watch your own movies and shows	Media HDD
Navidrome	Listen to your own music collection	Application SSD
Nextcloud	Store and access other personal files	Media HDD; app/database on SSD
Seerr	Request movies/shows through configured sources	Application state on SSD

After setup: your phone can reach these five services from home or away through Tailscale. Their files stay on your server. Tailscale supplies the private connection; it is not a storage account or a backup destination.

What “ready” means

The current release has automated test coverage, but this guide is not evidence of a completed installation on your future hardware. The first real installation still needs the reboot, phone, backup and restore tests in section 18. A simulated screen or example address is an explanation, not proof that your server is running.

Guide map by section: preparation 1-3; installation/accounts 4-6; remote connection 7-11; phone apps and mobile test 12-14; backups/daily use 15-17; completion and help 18-20.

1. Who does what?

The installer handles repeatable setup. You supply physical choices, personal authorizations and the final real-device tests.

Stage	Installer does	You still do
Before installation	Checks host requirements when launched	Install Linux; prepare formatted drives; know your sudo password
Installation	Installs missing packages; deploys selected containers, services and schedules	Confirm exact disks, user IDs, network and account choices
App accounts	Creates or verifies supported app accounts; connects supported local APIs	Choose generated/custom credentials; check VERIFIED results
Content	Configures libraries and supported connectors	Choose lawful content sources and enter any provider credentials
Remote access	Installs Tailscale and sets up private app routes after authorization	Authorize server and phone; enable DNS/HTTPS; test mobile data
Phone apps	Server is ready to accept authenticated clients	Install apps, enter server URLs, sign in, grant photo permissions
Ongoing use	Runs configured backup/update jobs and guarded recovery	Keep required disks connected, check capacity and failures, test restores

You do not need to keep a terminal open forever

For outside-Konsole tasks, the installer opens the relevant page/app and asks yes/no. No repeats its instructions and opens it again; yes also runs available checks. After a successful setup, system services and timers continue running without Konsole open. The server itself must stay powered on, connected to the internet and able to reach its storage. A shutdown or home internet outage makes remote access unavailable.

Optional work is separate from basic phone access

Steam/TV/controller setup, Samba file sharing, notification destinations and external heartbeat services have their own choices or logins. They do not replace Tailscale or app accounts. Leave their status visibly pending if you are not configuring them yet; do not treat pending as complete.

2. Prepare the PC and drives

Do this on the server before running the installer. Use the model, serial number and existing contents to identify a disk; capacity alone is not enough.

Planned device	Purpose	Installed path
256GB OS SSD	Existing Linux installation	Your OS filesystem
1TB application SSD	Containers, databases, photos, music	/srv/nekoserver
12TB media HDD	Movies, shows, cloud files, backups	/mnt/nekoserver-media
512GB external disk	Independent backup copy	/mnt/nekoserver-photos

Prepare supported, formatted filesystems on the intended data drives using the OS disk tools. Formatting erases data: first verify the device and keep copies of anything important. The NekoServer installer does not format them for you. The app SSD and HDD must be separate physical drives.

1. Log in as your normal Linux desktop user. Open **Konsole**, the terminal application where commands are typed.

2. Check your user and group IDs, then set the intended time zone. These examples use your planned Asia/Jerusalem schedule:

```
id -u
id -g
sudo timedatectl set-timezone Asia/Jerusalem
timedatectl status
```

Record the two numeric IDs. Many systems use 1000, but do not assume yours does. **sudo** means “run with administrator privileges.” Its password prompt asks for your Linux account password; nothing appears as you type. This is separate from the new server app passwords.

3. Have working internet, DNS, current time, sudo access and usable Arch package repositories. Steam needs the host multilib repository enabled if you want that optional feature.

Expected result: Linux is running normally, each intended disk is identified, and you know the desktop username and numeric IDs. If you cannot confidently identify a disk, resolve that before confirming storage prompts.

3. Extract and check the configuration

Do this in Konsole on the server. These commands assume the ZIP extracts directly into Downloads. Adapt the folder if your archive tool creates an extra directory.

```
cd "$HOME/Downloads/nekoserver-v0.21.0"
cp config.json my-config.json
chmod 600 my-config.json
```

The **configuration** is the JSON file containing your setup choices. Keep the original sample and work in my-config.json. Open that working copy in your text editor before installation. JSON needs matching braces and commas; keep numbers as numbers and text inside quotation marks.

1. Set **media.uid** and **media.gid** to the numeric IDs you recorded. Check **media.timezone** and other enabled schedule time zones match your intention.

2. If using TV features, set **tv.user** to your existing Linux desktop username. This is not a new app username.

3. For a fresh setup, leave the unconfirmed drive UUID fields empty so the installer asks you to identify them. A **UUID** is the identifier of a particular filesystem, not its friendly disk name.

```
bash install.sh --profile full --check \
  --config "$PWD/my-config.json"
bash install.sh --profile full --dry-run \
  --config "$PWD/my-config.json"
```

check inspects requirements. **dry-run** previews planned work. Neither installs packages or changes the system. If Python is missing, these previews may stop with that explanation; the apply launcher on the next page handles missing base dependencies.

Copy complete command blocks, including the backslash at the end of a continued line. Do not type a terminal prompt symbol such as $. **$PWD** expands to your current folder; it is why you first change into the extracted release directory.

Expected result: the configuration parses and the preview shows the intended profile. Resolve wrong paths or configuration errors before applying.

4. Run the installer and answer prompts

Stay at the server for the first run. Downloading images and starting databases can take time; keep the terminal open until the run finishes.

```
sudo bash install.sh --profile full --apply \
  --config "$PWD/my-config.json"
```

The launcher checks base tools first and installs missing prerequisites through pacman when needed. Selected modules then install their dependencies, such as Docker/Compose, Caddy, Samba and disk tools. The bundle includes its verified Rust executable; you do not need to compile Rust. This is not a full pacman/yay operating-system upgrade.

Storage prompts

Read the actual model, serial, filesystem and UUID in the inventory. Choose the dedicated app SSD, then the media HDD, then the intended external backup filesystem when asked. Type the exact UUID requested. Example identifiers in this guide must never be entered on your PC.

The installer saves confirmed choices, sets up the mounts and startup guards, and preserves a release/configuration copy on the app SSD. Wrong or missing storage is meant to stop dependent work instead of filling the OS SSD by accident.

Network and security prompts

For the home gateway, select the real private home-network interface/subnet. A router DHCP reservation helps keep its home IP stable; this is a router setting you make yourself. The installer opens the selected apps and explains their login/access checks. Answer **yes** after checking them, or **no** to reopen the pages and repeat the instructions. Available account checks also run before access is enabled.

If the installer stops or reports partial completion

Read the reported reason. success means that module completed; needs_setup means a personal action remains; failed means an error occurred; blocked means a prerequisite did not complete. Other independent modules may still finish.

```
sudo bash install.sh --retry-failed --apply \
  --config "$PWD/my-config.json"
```

Retry after fixing the cause. Do not replace your confirmed configuration with the empty sample. The working config becomes private/root-owned after saving; use sudo when access is required.

5. Choose and retrieve app logins

The full installer prompts for credentials and attempts supported account setup. You do not need to invent database passwords or manually wire every local connector.

1. Choose generated credentials for all supported accounts, or choose individually for each app. For a custom password, enter it twice when prompted; input is hidden.

2. Generated credentials use cryptographically secure randomness and a distinct 32-character password. Custom usernames start with an ASCII letter and use letters, numbers, underscores or hyphens, up to 32 characters. New custom passwords are 12-128 characters.

3. Immich and Seerr use email-shaped logins. A generated @server.invalid identifier works locally but cannot receive password-recovery mail. Configure a real address/mail service separately if you need email recovery.

Navidrome has two roles: an **administrator** for settings and a separate **listener** for normal music listening. Existing app data is handled by checking the current credentials, not silently resetting the app.

Read your private account inventory on the server

```
sudo less \
  /srv/nekoserver/secrets/accounts/APP-CREDENTIALS.txt
```

Use the arrow keys to scroll and **q** to exit. Every section names the app, its username, password and verification status. **Only VERIFIED means the installer successfully checked that login.** If automatic setup cannot finish an account, the installer opens its page with instructions. No reopens it; yes retries the authenticated check. A cancelled or unverified account stays pending.

The file is plaintext and protected as a root-only file in a private directory. Keep its contents private; a password manager is a convenient place to retain the logins. The file is an installation record, not a password manager that learns later changes made inside an app.

Two independent kinds of login

Tailscale sign-in: authorizes your phone or server to join your private device network. **App sign-in:** authorizes access to photos, files, music or video inside that application. Logging into Tailscale does not automatically log you into all the apps. Never enter your identity-provider password into an Immich/Jellyfin login form.

6. What is connected automatically?

The local server connections and your phone connections are different parts of setup. Both need to work.

Connection	Automatic part	Still needs your choice/test
Seerr to Jellyfin	Initial account/link setup uses verified local credentials	Sign in and verify the library/request workflow
Seerr to Sonarr/Radarr	Supported onboarding configures request targets	Choose providers/sources and complete a real authorized request
Media managers to Prowlarr/qBittorrent	Supported connectors share API/download settings	Supply working indexers/provider credentials and test import
Immich to its database/storage	Containers, paths and internal secrets configured	Choose phone albums; upload and view a photo
Nextcloud to its database/files	App, database, data path and initial admin configured	Sign in; upload/download; test its phone client
Navidrome to music library	SSD library, account setup and listener configuration	Add your own music and test playback
Phone to all five apps	Private routes created by remote-access profile	Authorize Tailscale, copy URLs and sign into phone apps

App-specific APIs can change. Unsupported responses or missing personal choices remain pending instead of being declared complete. A configured connector is not proof that an actual file downloaded, imported and played successfully.

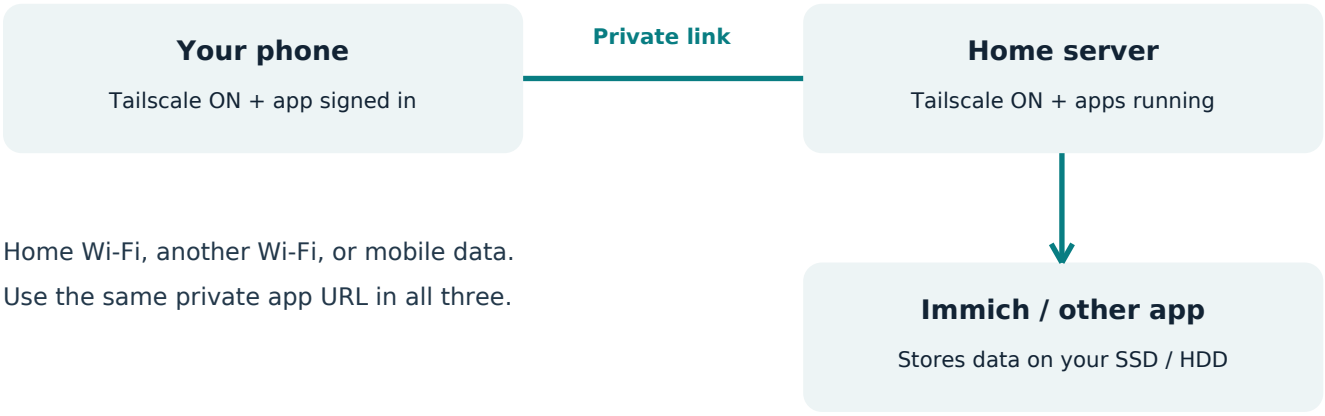
There is no bundled streaming catalogue

Jellyfin and Navidrome serve your own media. Navidrome can provide a familiar music-library experience through compatible phone clients, but it does not include Apple Music subscriptions or its commercial catalogue. Seerr is a request interface; it still needs the selected downstream services and content sources.

Expected result: local account checks pass, library paths are correct, and each service can perform a small real task before you start remote setup.

7. How the phone connection works

Tailscale makes your authorized phone and home server reachable through a private network, even when they use different internet connections.



Your phone first joins your private Tailscale network, called a **tailnet**. The server joins the same tailnet. The phone then opens the server's private HTTPS address, and the server sends that request to the selected local app. The app checks its own username and password before returning your files.

What stays the same when you leave home?

Use the same generated **https://...ts.net:port/** address at home and away, with Tailscale connected. You do not need to switch between a home IP address and an outside address. A port, such as 8445, selects which app receives the connection.

What must remain on?

The home server, its storage and home internet must be available. Your phone needs internet and an active Tailscale connection. Tailscale may request reauthentication later; server device expiry and access policy should be reviewed for unattended use. It is not a promise of permanent access without maintenance.

What this setup does not require

You do not need router port forwarding, a public IP address, a public Funnel route, a subnet router or an exit node for these five apps. The automation uses private Tailscale Serve routes. It does not expose the media administration tools or the local dashboard through those routes.

The private Tailscale HTTPS path uses its own certificate handling. You do not install the home gateway's private Caddy certificate on the phone when you use these .ts.net URLs.

8. Guided Tailscale sign-in

After the base apps work, start this separate profile in Konsole on the server. It now guides the browser and phone steps in one interactive run.

1. Point a terminal variable at the preserved release. Run this assignment again whenever you open a new terminal:

```
NEKO_RELEASE=/srv/nekoserver/installer/releases/v0.21.0
```

2. Run remote-access setup with the confirmed configuration:

```
sudo bash "$NEKO_RELEASE/install.sh" \
  --profile remote_access --apply \
  --config "$NEKO_RELEASE/config.json"
```

3. The installer installs/enables Tailscale and, if sign-in is needed, starts its authorization flow and opens the browser link. Sign in using the account/tailnet you want to use for your own phone. Approve the server, including admin device approval if your tailnet requires it.

4. Return to Konsole. At **Have you completed this step successfully? [yes/no]**, type yes after approving the server. Type no if you closed the page or need the instructions again; it opens the step again.

The installer checks that the local Tailscale backend is actually connected. If you answer yes but the check fails, it repeats the step. An already connected server is not forced to sign in again.

If a window does not appear

The installer prints the address for you to open manually. Browser/app launch runs as your normal desktop user, never as root. If the CLI cannot supply a link, the terminal explains the manual fallback below. In a normal guided run you do not need this separate command:

```
sudo tailscale up
```

Ctrl+C or unavailable input leaves the step pending and preserves saved choices. Reapply the same profile interactively to resume. Read-only and background runs do not open windows or wait for answers.

9. Guided HTTPS and app links

After server approval, the same remote-access run proceeds to private HTTPS setup. You normally do not need a second command or profile run.

1. The installer opens Tailscale DNS settings at https://console.tailscale.com/admin/dns. Select the correct tailnet, enable **MagicDNS** and **HTTPS certificates**. Read the notice about machine/tailnet names in public certificate records; it does not publish your files.

2. Return to Konsole and answer yes. No reopens DNS settings and repeats the explanation. After yes, the installer attempts the private app routes; an unavailable prerequisite keeps the step pending/repeating instead of falsely passing.

If asked to verify app security, it opens the selected app pages, explains the login/signup/access checks and asks yes/no. With managed accounts enabled, the supported authenticated logins must pass too. Do not clear unrelated routes or enable public Funnel to bypass a conflict.

3. The installer prints the actual URLs while guiding your phone setup. You can also read them later under the **links** section of this report:

```
sudo python3 -m json.tool \
  /srv/nekoserver/reports/gateway-tailnet.json
```

Copy your actual URLs into private setup notes. An example is https://nekoserver.tail12345.ts.net:8445/ for Immich, but your hostname differs. Do not enter this placeholder on your phone.

What does confirmation prove?

Route configuration is checked on the server. Phone use is checked by you, with explicit instructions and yes/no retries for each selected service and the mobile-data test. The report labels that evidence user_confirmed; it is not an independent automated probe of the phone.

Nextcloud after a full-profile rerun

The gateway applied last sets Nextcloud’s primary address. Remote setup makes the Tailscale URL primary. If you later rerun full, finish by reapplying remote_access so Nextcloud uses the intended Tailscale address again.

10. Connect your phone to Tailscale

Do this on the phone. Authorizing the server alone cannot make an unconnected phone reach private server addresses.

1. Follow the installer’s phone instructions and yes/no prompts. Install the official **Tailscale** app from your phone’s app store. Open it and begin setup.

2. Allow the requested VPN configuration. On an iPhone, approve the system dialog using the phone’s usual authorization method. This grants Tailscale permission to create the private connection.

3. For your own phone, sign in with the **same Tailscale account/tailnet** used for the server. If you manage multiple tailnets, check that the selected one contains the home server. Complete any device approval required by that tailnet.

4. Turn the connection on and confirm the app shows it is connected. You should be able to identify the server among your tailnet devices. Keep Tailscale connected while using these private app URLs.

5. Still on home Wi-Fi, open your phone browser and paste the **actual Immich URL** from the generated report, including https:// and :8445. A login page should load without a certificate warning. Log in with the Immich account, not the Tailscale account.

What is automatic after this?

The phone and server use the same private addresses whether the phone is on home Wi-Fi, another Wi-Fi network or mobile data. You do not manually reconnect each server app to your router. If Tailscale is disconnected, signed out or blocked by policy, those private addresses will not be reachable remotely.

Other people’s phones

Do not give someone your Tailscale owner password. Additional people need an intentional invitation/device-access policy and appropriate app accounts. Joining a tailnet and being permitted to see all your personal photos are separate decisions.

Expected result: the phone browser reaches an app’s real login page over the private HTTPS link. If it does not, use section 19 before changing passwords or reinstalling apps.

11. Enter the right URL in each app

The hostname is the same across the services; the port selects the app. Copy the complete URLs from your report so you do not have to construct them.

Phone app / browser	URL ending on your server	Which login?
Jellyfin client	:8443/	Jellyfin account
Seerr in browser	:8444/	Seerr account / configured Jellyfin sign-in
Immich client	:8445/	Immich email + password
Navidrome-compatible music client	:8446/	Navidrome listener account
Nextcloud client	:8447/	Nextcloud account

For example, the Immich server field might contain **https://nekoserver.tail12345.ts.net:8445/**. That is an illustration only. Use the hostname in your own links report. Start with the complete base URL; do not guess extra /api paths unless the particular client explicitly requires one.

Three easy address mistakes

1. localhost or 127.0.0.1: on a phone this means the phone itself. The server's local addresses are only useful from the server.

2. 192.168.x.x or another private home IP: this guide's remote app routes use .ts.net addresses. A home-LAN address is a separate connection path and is not made remotely reachable by these five Serve routes.

3. Missing port or wrong protocol: include https:// and the exact :844x port. The home portal on :8442 is not a remote Tailscale portal in this setup.

Use browser first, native app second

If an app fails to connect, open that same complete URL in the phone browser. If no login page loads, fix connectivity or HTTPS first. If the browser works but the native app fails, check the native app's server field, account type and credentials.

Expected result: each desired phone app saves its server address and account. You normally enter this once, unless its saved settings, credentials or server name change.

12. Set up photos and personal files

Do this after the browser connection test. Phone permissions are intentionally under your control; the server installer cannot grant them remotely.

Immich: first photo backup

1. Install the official Immich phone app. Enter your actual :8445 server URL and sign in with the verified Immich email/password.

2. Grant access to the photos you intend to back up. If you grant limited access, Immich can only see that selection. Choose the albums/folders you want included in its backup settings.

3. Choose whether uploads may use mobile data or should wait for Wi-Fi. Start a small backup and keep the app open until the test finishes. For the first large library, home Wi-Fi avoids unnecessary mobile-data use.

4. Open the uploaded photo through the server and verify a video plays too. The phone saying “selected” is not the same as the server storing the file. Background work depends on phone permissions, power settings and OS scheduling; do not assume every photo is instantly uploaded when the app is closed.

Immich writes live data to the SSD. Separate server backup jobs later copy the required data to backup storage. A phone upload and an independent backup are different events. Keep phone originals until you have verified the backup/restore path.

Nextcloud: first file

1. Install the official Nextcloud client. Enter the actual :8447 URL. Follow its browser/app login approval flow using the Nextcloud account.

2. Upload a harmless test document. Check it in the web interface, then download/open it from the phone. This verifies both directions.

3. Select any optional Nextcloud auto-upload or offline-folder settings yourself. Immich photo backup does not automatically turn on Nextcloud photo upload; enabling both may create two copies by choice.

Nextcloud user files live on the HDD. Add/change them through Nextcloud’s supported clients or web interface. Writing directly into its managed files directory through Samba can bypass its database and is not the normal upload workflow.

13. Set up music, video and requests

You can test these services with small files you already own before configuring a large collection or any download providers.

Music with Navidrome

1. Put a small owned music sample into the configured SSD music library using the intended import/share path. Confirm Navidrome sees the album after its scan.

2. Choose a Navidrome/Subsonic-compatible client for your phone. Navidrome is the server; it does not itself supply every phone's native music app.

3. Enter your actual :8446 URL and the **listener** account from the private inventory. Play a full track, seek forward and test another track. Reserve the separate admin account for administration.

Music files stay on SSD. The chosen policy has **no automatic backup of raw music files**. Settings, accounts and supported app state are protected separately. Keep another source copy yourself if you cannot replace your music.

Jellyfin for movies and shows

1. Confirm an owned sample appears in the configured HDD video library. Use Jellyfin's native client or browser with the actual :8443 URL and Jellyfin account.

2. Play a sample, seek, and check audio/subtitles. Remote quality depends on home upload bandwidth, phone connection and any needed transcoding; successful home playback alone does not prove remote 4K playback.

Seerr for requests

1. Open the actual :8444 URL in a phone browser and sign in. Confirm the Jellyfin library is recognized.

2. Finish selecting provider/indexer sources and their required credentials in the supported setup. Then test one permitted request through request, download, import and playback.

A request being accepted is only the first stage. Do not call the chain complete until the resulting item actually appears and plays in Jellyfin. Movies/shows are HDD-only and are intentionally excluded from raw media backups.

14. Prove it works outside home

This is the test that answers “will my phone work away from my Wi-Fi?” Do it before you depend on remote access.

1. Leave the server running at home with its drives and internet connected. Verify Tailscale is connected on both server and phone.

2. On the phone, turn **Wi-Fi off** and ensure mobile data is enabled. Confirm it is really using the cellular connection.

3. In the phone browser, reload the real Immich .ts.net URL and sign in. Open a photo that needs fetching from the server; a previously cached thumbnail is not proof of a working connection.

4. Open Nextcloud and fetch the test document. Play a music track from Navidrome, then play a short video from Jellyfin. Open Seerr if you plan to use requests remotely.

5. If you permit cellular photo uploads, create one harmless new photo and confirm Immich receives it. If you chose Wi-Fi-only backup, a delayed mobile upload is expected; test viewing on mobile and uploading later on Wi-Fi instead.

6. As a private-access check, disconnect Tailscale on the phone and attempt a fresh server fetch while still on mobile data. It should fail to reach these private routes. Reconnect Tailscale and repeat; it should work. Cached/offline media can still display, so test a fresh request.

A passing test means

Your phone reaches the intended server from another network, the app accepts the right account, and the relevant upload/playback/file operation works. Record which tests passed and which remain pending. Server-side configuration alone cannot supply this evidence.

Normal use afterward

Keep using the saved .ts.net URLs at home and away. Leave Tailscale connected while using the services. The apps may keep their sessions signed in, but expired sessions, password changes or Tailscale reauthentication can require attention later.

Mobile playback and uploads consume data. Remote streaming speed is limited by the slowest relevant connection, including your home internet’s upload speed.

15. Where data and backups go

These are the intended policies, not a claim that every file is backed up everywhere. Capacity must be sufficient on each destination.

Data	Live location	Independent copy policy
Immich photos/videos + database	SSD	Weekly HDD originals + database preservation; weekly external originals. Monthly complete recovery bundle to SSD/HDD/external.
Nextcloud user files	HDD; app/database on SSD	Monthly full recovery copy including files, database and configuration; external monthly, not weekly.
Music audio files	SSD	No automatic raw-music backup.
Movies/shows	HDD	No automatic raw-video backup.
Supported app configuration/accounts/state	SSD	Daily app-state protection plus monthly recovery bundle; private inventory included when present.

Weekly jobs

Sunday 02:15, Asia/Jerusalem: Immich HDD originals archive and logical database preservation. **Sunday 03:00:** external photo-originals copy, waiting for the active local photo job when needed. The external drive must be connected and recognized.

An external originals copy alone is not a full Immich restoration. A full recovery also needs the matching database/configuration. Those are included in the complete monthly bundle. This distinction matters if a disaster occurs between monthly full backups.

Monthly job

First day of the month at 02:30, Asia/Jerusalem: full cold app/Immich/Nextcloud backup, exact old app images, verified copies and an isolated restore rehearsal precede supported updates. On failure, recovery uses matching old application images and old data.

Daily app-state work is scheduled at 04:15, and selected host/Steam/Samba state at 05:15 using the host’s local time. Keep that host time zone consistent with your plan. These jobs can access disks even though the removed hourly media checkup stays disabled.

16. What still needs care over time?

Automation reduces repeated work. It cannot replace a connected backup drive, sufficient capacity, working internet or a tested recovery procedure.

Keep the external disk available

A missing, wrong or full external drive blocks the guarded monthly update path. Connect the correct drive before the scheduled run and check the result. Do not assume a missed run succeeded or will immediately catch up when you plug in a disk; inspect the recorded status and follow the maintenance instructions for a retry.

Watch capacity

Your planned 512GB external disk may eventually be too small for a full backup of photos, Nextcloud and application state. A 1TB SSD or 12TB HDD does not make that external copy fit. The automation checks space; it does not buy a drive, grow filesystems or automatically delete old generations for you.

Understand updates and rollback

Monthly guarded updates are for the supported application workflow. They do not automatically perform a complete Arch/pacman/yay upgrade, reboot the PC or migrate across arbitrary major app versions. Host OS maintenance and major migrations remain deliberate tasks.

A rollback is useful only if its backup contains the required matching data and images. Do not delete backup generations merely because the new app launched once. Recovery should be rehearsed on an isolated destination; restoring over live data is a separate, destructive action.

Know when a job failed

Notification destinations and any external heartbeat receiver must be configured and tested by you. Until you have received a real test notification, do not rely on alerts. A quiet phone is not proof that all backups succeeded.

Review access as devices change

Remove lost phones from Tailscale, review which people/devices can reach the server, and handle requests for reauthentication. App passwords and Tailscale device access are separate controls. Updating an app password in its UI does not automatically update every phone client or the original credential inventory.

17. Daily use, reboot and HDD idle

The intended routine is to use the phone apps normally, let scheduled jobs run at night, and check any reported failure.

Sleep and the HDD are different

The server is configured to stay awake. The guarded SATA HDD idle policy uses a 60-minute timeout where the hardware supports it. A sleeping HDD can wake when a file is requested; the whole PC must not suspend if you expect remote access.

There is no routine hourly media API checkup in this policy. However, Nextcloud access, Jellyfin/library scans, downloads/seeding, backup jobs and other legitimate work can still wake the HDD. No policy can promise it will spin down if applications continue accessing it.

RAM use is not automatically a leak

Linux uses otherwise available memory as filesystem cache and can reclaim it when applications need RAM. A high “used” figure alone is not evidence that the OS is filling RAM permanently. This setup does not schedule cache-clearing commands. Investigate repeated out-of-memory events or growing application memory, rather than flushing useful cache on a timer.

After a restart

1. Give the server time to mount its required disks and start services. Missing storage should block dependent startup rather than send data onto the OS SSD.

2. Check that the server appears connected in Tailscale. From the phone, open a fresh photo/file and play a track.

3. If Nextcloud redirects to a home IP after you reran the full installer, reapply remote_access as explained in section 9.

Home TV and local HTTPS

Home clients using the separate Caddy LAN addresses must trust its private root certificate and use the correct server address. The phone route in this guide uses Tailscale HTTPS, so that private certificate step is unnecessary there. Steam/TV use still requires the relevant external sign-ins, controller and HDMI tests.

18. The final “what am I left to do?” list

Do not mark the server complete just because the terminal reached its final line. Tick these only after observing the result.

Required for the intended basic server	Evidence to record
Correct disks and mounts	Confirmed UUIDs and enough space on all required destinations
App account setup	Every app you need says VERIFIED; you can sign in
Local service connection	Real photo upload, file round-trip, music/video playback
Remote server authorization	Tailscale connected; private gateway report includes actual links
Phone connection	Tailscale VPN authorized; correct tailnet; saved app URLs
Phone permissions	Chosen albums visible; intended Wi-Fi/mobile upload policy works
Outside-home access	Wi-Fi-off/mobile-data tests pass for every app you need
Backup and recovery	Real backup verified; restore rehearsed to a separate isolated target
Reboot behavior	Required disks mount; apps and Tailscale return; phone test passes
Operational follow-up	Capacity/retention plan; alert delivery tested if used

Run the server-side acceptance helper too

```
NEKO_RELEASE=/srv/nekoserver/installer/releases/v0.21.0
sudo python3 "$NEKO_RELEASE/acceptance.py" \
  --profile full --save --manual \
  --config "$NEKO_RELEASE/config.json"
```

With --manual, it opens relevant pages/apps and guides witnessed tests with the same yes/no retry flow. Read its output and saved evidence. This helper supplements the physical phone/reboot/restore tests; it cannot personally see that your phone played a video or that a restored file is the right one.

Additional features: finish provider/indexer choices for automated requests, external notifications for alerts, Samba for network shares, and Steam/controller/TV setup if you want them. Record anything deferred as pending. Bazzite migration is separate work.

19. If something does not work

Start with the failing layer. Reinstalling everything or changing all passwords rarely identifies the actual cause.

Symptom	Check in this order
No .ts.net page loads on phone	Server power/internet; Tailscale connected on both devices; same tailnet; actual URL and port; HTTPS setup; intended access policy.
Report says needs_setup for Tailscale	On server run sudo tailscale up; finish browser approval; rerun remote_access.
Device is visible but app is unreachable	Device membership is not app readiness. Check private Serve routes and local app health; review policy for that port.
Login page opens, password rejected	Use that app's VERIFIED inventory entry. Confirm correct username/email and Navidrome listener/admin role. Tailscale password is different.
Phone browser works, native app fails	Recheck complete server field, HTTPS, port and selected account. Remove mistaken extra paths; follow that client's login flow.
Immich does not upload new photos	Check album selection, full/limited photo permission, backup enabled, Wi-Fi/mobile policy, phone background restrictions and server free space.
Nextcloud redirects to 192.168...	The LAN gateway may have been applied last. Rerun remote_access with the confirmed config to restore the Tailscale primary URL.
Monthly update did not proceed	Read its failure reason. Correct external drive identity/connection/free space and backup/rehearsal prerequisites before retrying.

Useful read-only checks on the server

```
tailscale status
tailscale serve status
sudo python3 -m json.tool \
  /srv/nekoserver/reports/gateway-tailnet.json
```

Do not disable HTTPS checks, turn on public Funnel, open router ports or clear unrelated routes as a shortcut. The expected path here is the private, authorized one.

20. Keep this reference

Your private working configuration, real app URLs and account inventory are the useful handover. The ZIP contains code and examples, not your finished server passwords.

Item	Location / reminder
Preserved release	/srv/nekoserver/installer/releases/v0.21.0
Confirmed configuration	config.json inside that preserved release; keep it private
Account inventory	/srv/nekoserver/secrets/accounts/APP-CREDENTIALS.txt
Private app links	/srv/nekoserver/reports/gateway-tailnet.json, links section
App SSD mount	/srv/nekoserver
Media HDD mount	/mnt/nekoserver-media
External backup mount	/mnt/nekoserver-photos

Write these down privately

Server Tailscale hostname: _____
Tailscale account/tailnet: _____
Password-manager location: _____
First successful mobile-data test: _____
First successful isolated restore test: _____
Features still pending: _____

Official references for the remote setup

Install Tailscale on iOS
https://tailscale.com/docs/install/ios

Private app sharing with Tailscale Serve
https://tailscale.com/docs/features/tailscale-serve

MagicDNS / HTTPS certificate setup
https://tailscale.com/docs/how-to/set-up-https-certificates

Prepared 18 October 2026 for NekoServer v0.21.0. Commands, route ports and account behavior were checked against that release’s installer and gateway implementation. Diagrams and example addresses explain the intended setup; they are not screenshots of your future server.